

Sohil Ramachandra

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[Website](#) – [GitHub](#) – [LinkedIn](#)

EDUCATION

University of Michigan

Ann Arbor, MI

Master of Engineering in Computer Science

August 2024-May 2025

GPA: 3.89/4.00

Bachelor of Engineering in Computer Science, Minor in Mathematics

August 2021-May 2024

GPA: 3.75/4.00

Relevant Coursework: Operating Systems, Machine Learning, Compilers, Web Systems, Database Management Systems, Data Structures and Algorithms, Cryptography

SKILLS

Languages: Python, C++, C, Java, Go, SQL, JavaScript, CSS, HTML

Tools: Git, Docker, Visual Studio, GDB, AWS, Linux

Frameworks/Libraries: React, TensorFlow, PyTorch, scikit-learn, Pandas, Boost Threads, Matplotlib, LLVM

WORK EXPERIENCE

Aptiv

Troy, MI

Cybersecurity Engineer

June 2025 - Present

- Lead threat modeling and risk assessment across software and hardware interfaces, ensuring alignment with cybersecurity standards.
- Translate system-level cybersecurity requirements into actionable hardware/software specifications in collaboration with cross-functional product teams.

Software Engineering Intern (Cybersecurity Team)

May 2024 - August 2024

- Built a Python automation tool that integrated with the Finite State API (an external security service) to handle file uploads, scan the retrieved report, and parse results, saving engineers ~5 hours per week.
- Designed and implemented a recursive algorithm to collect all necessary source files for scanning, ensuring comprehensive security coverage.
- Created a GUI that allowed users to input project names and API keys and view scan results, reducing onboarding time and boosting tool adoption.

Software Engineering Intern (Quality Assurance Team)

May 2023 - August 2023

- Developed an automated system in Python to streamline the production of the Quality Assurance Matrix (QAM) by integrating data from engineering drawings and inputs, which significantly boosted productivity by eliminating manual compilation efforts.
- Designed and implemented a user-friendly GUI to facilitate easy file input from users' computers, optimizing the accessibility and usability of the system.

PROJECT EXPERIENCE

Network File Server

January 2024 - April 2024

University of Michigan

Ann Arbor, MI

- Engineered a secure, multi-threaded network file server in C++ from scratch, supporting create/read/append/delete operations via custom client-server protocol over TCP sockets.
- Designed concurrency-safe file operations using Boost threads and reader-writer locks, enabling non-blocking parallel access across multiple clients.
- Built a custom on-disk file system with inodes and block management, including logic for consistent crash recovery and atomic operations.

Machine Learning for Register Allocation

August 2024 - December 2024

University of Michigan

Ann Arbor, MI

- Designed a transformer-based model in Python to translate LLVM MIR code with virtual registers into optimized physical register mappings, achieving 89% accuracy on validation data.
- Engineered a full data pipeline using LLVM passes and custom tokenization to process 8,000+ functions, enabling reliable training and correctness-preserving inference.
- Optimized training efficiency with PyTorch's mixed-precision, batching, and token filtering, which reduced runtime and improved performance on long function sequences.